



Nursing Practice Guideline

Percutaneous Nephrostomy Catheter

Reference #: NPG24

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to management of Percutaneous Nephrostomy Catheter (Tube). It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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1. Principles

A nephrostomy catheter is a narrow-gauge pigtail drain inserted percutaneously into the renal pelvis for the purpose of draining urine.¹ Nephrostomy catheters' are inserted in the Radiology Department or the Operating Theatre to provide temporary or permanent urinary drainage following a procedure, or to relieve ureteric obstruction.²

Indications for inserting a nephrostomy catheter include:

- To enable removal of renal calculus
- To decompress an obstructed system and to maintain or improve renal function following ureteric obstruction
- To obtain access to the renal pelvis for radiological procedures e.g. insertion of an ante grade stent.² or for delivery of medications.

Observe the five moments for hand hygiene and aseptic non-touch technique. Follow the risk assessment outlined SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy. Appropriate Personal Protective Equipment (PPE) must be worn to prevent contamination and mucosal or conjunctival splash injuries.

Patient identification must be checked when matching patient's identity to care, therapy and services.

2. Management of Nephrostomy Catheter

2.1 Medical Requirements

Urea, electrolytes and creatinine must be ordered and monitored by Medical officer. ²

2.2 Safety Information

Post insertion if there is sudden increase, decrease or cessation in nephrostomy catheter drainage - notify Shift Co-ordinator and Medical staff.

- No urine output may indicate catheter obstruction or catheter displacement.³
- Low urine output (less than 30mls per hour for 2 consecutive hours ^{1,2}) may indicate dehydration or clinical deterioration.
- High volume outputs (polyuria of greater than 200mls per hour for 2 consecutive hours) may result from relief of urinary obstruction and requires careful monitoring of fluid balance and vital signs to prevent patient deterioration associated with rapid fluid loss. ¹ The post obstructive polyuria Departmental Guideline is available from the Urology Department to guide management and fluid replacement.

The nephrostomy catheter must be well secured at all times to prevent dislodgement. ^{1,4}

The nephrostomy catheter may or may not be anchored to the skin by sutures. An occlusive dressing must remain intact until otherwise ordered by Medical Officer.

Patients with a nephrostomy catheter are at risk of pyelonephritis and should be monitored for signs of infection / sepsis e.g. loin pain, elevated temperature, fever/chills, purulent urine or deterioration in vital signs.¹

A variety of nephrostomy tubes are in use; for specific product use and instructions refer to manufacturers guidelines.

2.3 Nursing Management

Post Insertion

- The nephrostomy catheter will be secured by a fixation device and will be connected to a sterile drainage bag (a leg bag is also appropriate).
- Complete post procedural observations as ordered by Interventionist.
- Ensure the nephrostomy catheter is secured below the hipline.^{1,2} Check connections, patency and that there are no kinks in the catheter tubing.^{1, 2,3}

Anchor the nephrostomy catheter to the skin with tape - e.g. Fixation device (Flexi Trak), Fixomul™. If a 2 or 3 way tap is in situ, ensure it is in the open position to facilitate free drainage.

Record the volume and nature of urine draining post insertion on a Fluid Balance Chart hourly for 4 hours, then as clinically indicated.^{1,2} Each urinary drainage route should be monitored separately and the overall total urine output calculated i.e. Left nephrostomy, right nephrostomy and urethral.¹

Note: it is normal for haematuria to be present immediately after insertion but this should decrease within 48 hours.²

- Ensure pain is well controlled and if necessary, administer prescribed analgesia.^{1,2}
- Monitor insertion site dressing for strike through / bleeding with post procedural observations.¹

Nephrostomy Catheter Care

- Check the urine drainage system each shift, for patency and drainage.^{1,2,3,4}
- Check the catheter for misplacement or kinking.¹⁻⁴
- Observe the dressing site for inflammation, signs of infection and leakage each shift.^{1,2}
- Assess patient for pain at insertion site, any alteration in skin integrity or any break in the seal of the dressing.^{1,2}
- Ensure the drainage bag is below the level of the kidney at all times. Do not place the drainage bag on the floor.^{1,2}
- The leg drainage bag should be worn with leg bag straps for comfort and ease of emptying. It can be attached to a straight drainage bag overnight⁵.
- The nephrostomy insertion site must be kept dry, protect whilst showering.^{1,2}
- The dressing, fixation device and drainage system should be changed weekly or more frequently if indicated.^{1,2}

Discharge Planning and Education

- Prior to discharge, ensure the catheter is secured with a fixation device as per Nephrostomy Catheter site dressing procedure.
- Provide a copy of patient information pamphlet 'Care of your nephrostomy tube' available from Urology Department and SCGH Urology ward and ensure they have understood the information.
- Educate patient and carers on management of the nephrostomy catheter.^{1,2}
 - Reinforce appropriate hand hygiene when handling the nephrostomy catheter, dressing and drainage system.²
 - Teach patient to empty leg bag when half full and how to attach / remove night drainage bag.
 - How to recognise signs and symptoms of nephrostomy catheter related infections.
 - How to contact community nursing services and when to seek medical advice.
- Supply patient with a spare leg bag and a two litre sterile night drainage bag.
- Refer to Silver Chain or community nursing service for ongoing care and monitoring.^{1,2}
- Ensure patient understands plan of follow-up care.¹

3. Nephrostomy Catheter Site Dressing

3.1. Safety Information

Refer to Section 1: Management of Nephrostomy Catheter

3.2. Nursing Management

- The dressing, fixation device and drainage system should be changed weekly or more frequently if indicated.^{1,2}

3.3. Procedural Information

Requirements

- Fixation device (universal Grip-Lok™ or Statlock™ of appropriate size e.g. 6-16Fr)
- Large transparent occlusive dressing (Tegaderm™)
- Normal saline 0.9%
- Dressing pack
- Adhesive tape (e.g. Fixomul™)
- Wet strength bag
- Sterile gloves
- 1 x skin barrier wipes
- 1 x adhesive remover
- Personal Protective Equipment (PPE)

Additional Requirements If Drainage System Change Required.

- Sterile Drainage bag (or Leg Bag)
- Extension tubing (male luer lock to catheter tip fitting)
- 3 way tap (only if flushing required)
- Needleless bung (only if flushing required)
- 1 x large 70% alcohol wipe

Procedure

1. Perform hand hygiene.
2. Clean the trolley using a combined detergent /disinfectant wipe, allow to dry.
3. Gather equipment.
4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique.
6. Perform hand hygiene.
7. Explain procedure to patient and position patient for ease of access to nephrostomy tube / catheter, placing an absorbent sheet underneath.
8. Perform hand hygiene and don nonsterile gloves^{3,4}.
9. Remove dressing and fixation device trying to minimise catheter manipulation during removal by using adhesive remover.
10. Perform hand hygiene, don sterile gloves^{3,4}.
11. If drainage system change is required, attach drainage bag to extension tubing.
12. If flushing is required, replace 2-way tap with 3-way tap and attach needleless bung to outlet port on 3-way tap.
13. Place sterile towel beneath nephrostomy tube.
14. Clean around entry site with normal saline 0.9% (take swabs if signs of infection present, notify medical team)⁴
15. Apply skin barrier wipe to surrounding skin.
16. Apply fixation device at or near insertion site^{1,4}.
17. Ensure nephrostomy catheter tubing is not kinked/twisted^{1,2}.

18. Position nephrostomy catheter tubing through the centre of the fixation device and secure.
19. Cover the insertion site and the fixation device with a sterile transparent occlusive dressing ensuring a good seal.
20. If drainage bag change is required, clean connection between nephrostomy tube and tap with a large 70% alcohol wipe, allow to dry for 30 seconds¹.
21. Disconnect nephrostomy tube from tap and replace with new drainage system
22. Ensure the nephrostomy catheter is secured below the hipline with tape e.g. Fixomul™^{1,2}.
23. Dispose of waste.
24. Remove PPE and perform hand hygiene.
25. Clean the trolley using a combined detergent disinfectant wipe.
26. Perform hand hygiene.
27. Document and record date of change in patient's notes and Nursing Care Plan.

4. Flushing of Nephrostomy Catheter

4.1. Medical Requirements

Flushing of a nephrostomy catheter is only to be performed on request from the Medical Officer and is prescribed on the WA Hospital Medication Chart (WAHMC).⁶

4.2. Safety Information

Refer to Section 1: Management of Nephrostomy Catheter

Following insertion of a nephrostomy tube, the patient may return to the ward with a 2 way tap in place. If flushing is required to maintain tube patency due to clots, blood or debris⁵, the 2 way tap needs to be changed to a 3 way tap to create a closed system suitable for flushing (see below).

The renal pelvis can only normally accommodate 4-8mls^{1,6}. The maximum volume of flushing solution should be 10mls unless otherwise ordered by Medical / Urology team.^{5,6}

If any excess pressure is required to flush or patient experiences pain, stop the procedure and notify Medical Officer immediately. ^{2,5,6}

4.3. Procedural Information

Changing 2 Way Tap to 3 Way Tap

Requirements

- Dressing pack
- 1 x Large 70% alcohol wipe
- 3-way tap
- Sterile gloves
- Needleless bung
- Sterile Drainage bag
- Extension tubing (male luer lock to catheter tip fitting)

Procedure

1. Perform hand hygiene
2. Clean trolley with combined detergent / disinfectant wipe. Allow to dry
3. Gather equipment for the procedure.

4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique
6. Perform hand hygiene.
7. Explain procedure to patient and position patient for ease of access to nephrostomy tube / catheter.
8. Perform hand hygiene and don sterile gloves³.
9. Attach drainage bag to extension tubing.
10. Attach extension tubing to 3-way tap and attach needleless bung to outlet port on 3-way tap.
11. Place sterile towel beneath nephrostomy tube³.
12. Clean connection between nephrostomy tube and 2-way tap with large 70% alcohol wipe. Allow to dry for 30 seconds¹.
13. Remove 2-way tap and existing drainage system from nephrostomy tube and replace with 3-way tap and new drainage system.
14. Ensure the nephrostomy catheter is secured below the hipline with tape e.g., Fixomul™.
15. Dispose of waste.
16. Remove PPE and perform hand hygiene.
17. Clean trolley with combined detergent /disinfectant wipe. Allow to dry.
18. Perform hand hygiene.

Flushing of Nephrostomy Catheter

Requirements

- Non sterile kidney dish
- 1 x large 70% alcohol wipe
- 1 x 10 mL syringe
- 1 x 10mL 0.9% sodium chloride for injection
- 1 x 18G drawing up needle
- Personal Protective Equipment (PPE)

Procedure

1. Assemble equipment in non-sterile kidney dish and prepare Normal saline flush as ordered.
2. Check patient identity against medication order.
3. Position patient for ease of access to nephrostomy catheter, place an absorbent sheet underneath.
4. Perform hand hygiene and don non sterile gloves.
5. Protect key parts using a non-touch technique. Scrub the needleless bung on the outlet port of 3 way tap with a large 70% alcohol wipe for 30 seconds. Allow to dry for 30 seconds^{5,6}.
6. Turn the 3 way tap off to the drainage bag and connect Normal saline flush to needleless bung using a non-touch technique.
7. Gently inject prescribed amount of normal saline (5-10mls) into nephrostomy tube – do not force or aspirate^{1,2,5,6}
8. Remove the syringe and turn the 3 way tap off to needleless bung to allow for gravity drainage into drainage bag.
9. Check that the entire system is patent and draining ^{1,5}.
10. Dispose of waste.
11. Remove PPE and perform hand hygiene.
12. Clean trolley using a combined detergent /disinfectant wipe. Allow to dry.

13. Perform hand hygiene.
14. Record administration on WAHMC, and if difficulty encountered notify shift coordinator, medical staff and record in patient notes.

5. To Obtain a Specimen of Urine

5.1 Medical Requirements

Complete CPOE specimen collection request form.

5.2 Safety Information

Refer to Section 1: Management of Nephrostomy Catheter.

All urine specimens must be collected from nephrostomy catheter by gravity. Do not use aspiration. ^{2,4}

5.3 Procedural Information

Requirements

- Dressing pack
- 2 x large 70% alcohol wipe
- Sterile gloves
- Sterile specimen jar
- Sterile kidney dish
- Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene.
2. Turn the 3 way tap off to nephrostomy catheter and give the patient a drink prior to the procedure (only if patient is not fasting).
3. Perform hand hygiene.
4. Clean trolley with combined detergent disinfectant wipe. Allow to dry.
5. Gather equipment for the procedure.
6. Perform hand hygiene and prepare field.
7. Open dressing pack and add equipment to sterile field using non-touch technique.
8. Perform hand hygiene.
9. Position patient for ease of access to nephrostomy catheter, place absorbent sheet underneath.
10. Perform hand hygiene and don sterile gloves³
11. Place dressing towel below tap.
12. Clean tubing and tap using a large 70% alcohol wipe.
13. Clean luer lock connection below the tap with 2nd large 70% alcohol wipe and allow to dry.
14. Hold the proximal end firmly to prevent kinking and separate extension tubing at connection below the tap.
15. Place distal end on the dressing towel.
16. Place sterile kidney dish on towel.
17. Turn 3-way tap off to needleless bung and allow urine to drip into kidney dish.
18. Turn 3-way tap off to nephrostomy catheter and reconnect tubing.
19. Turn the 3- way tap off to needleless bung to allow continuous drainage to resume.
20. Transfer urine to sterile specimen container.

21. Dispose of waste.
22. Remove PPE and perform hand hygiene.
23. Clean trolley with combined detergent /disinfectant wipe.
24. Perform hand hygiene
25. Record in patient notes

6. Removal of Percutaneous Nephrostomy Catheter

6.1 Medical Requirements

Removal of nephrostomy catheter must be documented in patient notes by Medical Officer.²

6.2 Safety Information

If the patient has a ureteric stent in situ, the removal of the percutaneous nephrostomy catheter is performed in the Radiology Department under imaging to prevent dislodgement of stent ^{2,7,8}

If there is difficulty removing the nephrostomy catheter, inform the medical team / Radiology or Urology Registrar.^{1,7,8}

6.3 Nursing Management

Ensure adequate analgesic (if required) is administered prior to removal of catheter.²

Tube removal is best performed from behind, with the patient sitting upright. If this is not possible, the patient may be positioned lying on their side and the procedure performed from behind ⁸.

6.4 Procedural Information

Requirements

- Normal Saline 0.9%
- Dressing pack
- 1 x adhesive remover
- Transparent occlusive dressing (Tegaderm™)
- 1 x skin barrier wipe
- Absorbent sheet
- Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene.
2. Clean the trolley with a combined detergent /disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique⁷.
6. Perform hand hygiene.
7. Explain procedure to patient and position patient for ease of access to nephrostomy catheter.
8. Perform hand hygiene and don PPE (non-sterile gloves).
9. Remove dressing and fixation device using adhesive solvent.
10. Perform hand hygiene.
11. Clean around the nephrostomy catheter with Normal saline 0.9%.
12. Apply skin prep to surrounding skin.

13. Unlock pigtail catheter as per manufacturers' instructions (see instructions below)^{1,7,8}
14. Don PPE (non-sterile gloves).
15. Gently remove tube while supporting the skin surrounding the drain. If any resistance is felt, inform the medical team / Radiology or Urology Registrar ^{2,7,8}.
16. Apply occlusive dressing to insertion site^{2,7,8}.
17. Dispose of waste.
18. Remove PPE and perform hand hygiene.
19. Clean the trolley with a combined detergent/ disinfectant wipe.
20. Perform hand hygiene.
21. Update Nursing Care Plan and document in the inpatient notes.
22. Continue to observe insertion site for ongoing drainage and leakage and educate patient to notify staff if leakage should occur. A pressure dressing over the site may be required.² A wound drainage bag may also be considered if leakage is excessive ⁸

A. Cook Medical Ultrathane® Nephrostomy set with Mac-Loc®

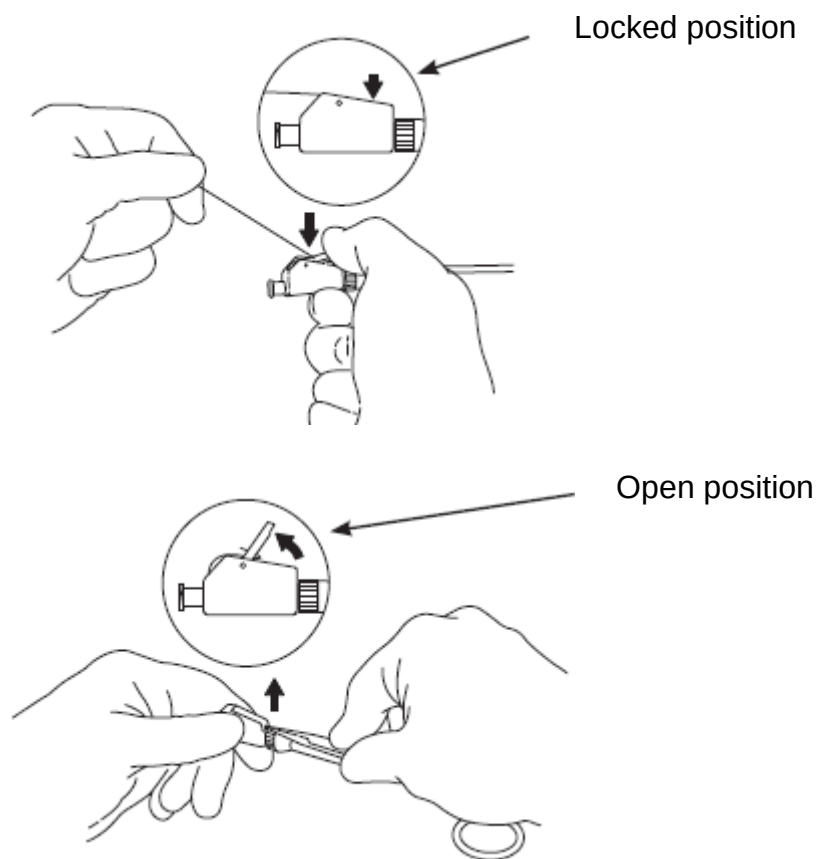


Figure 1: Mac-Loc® locking mechanism⁹

To unlock

- a) Whilst stabilising the Mac-Loc® catheter hub assembly with one hand, position a small blunt object (approximately the size and shape of a ball point pen or small forceps) into the Mac-Loc® release notch.
- b) Pry forward until the locking cam lever is free. ⁹

B. Merit Medical Resolve® Catheter

Unlocking with repositioning tool / key

- Align the opening of the round section of the repositioning tool in line with the handle of the suture locking mechanism
- Bring the flat back of the repositioning tool in line with the handle of the suture locking mechanism.
- Bring the flat back of the repositioning tool around the catheter.
- Gently squeeze together; this disengages the catheter locking mechanism.
- Rotate handle clockwise. Hub is now unlocked. ¹⁰

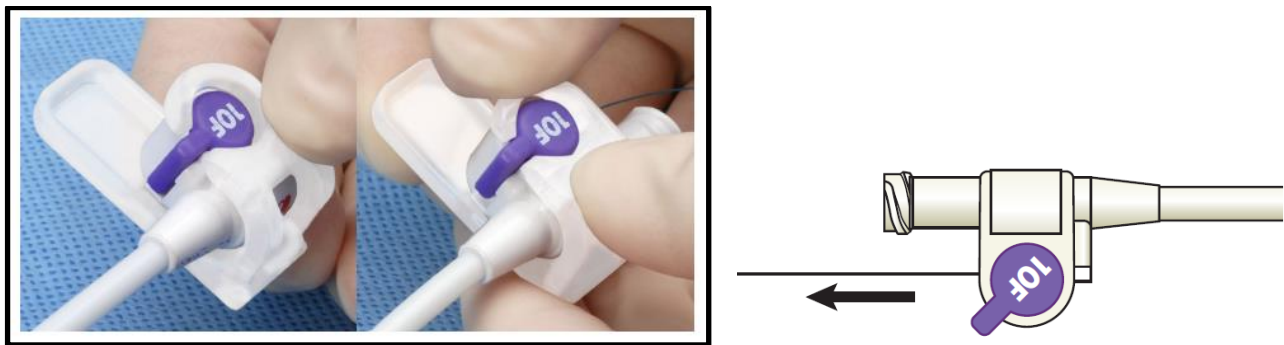


Figure 2 Repositioning Tool and unlocking¹⁰

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010
- Work Health and Safety Act 2020

Policy:

- NMHS Consent to Treatment Policy
- Health Mandatory Policy 0131/20 High Risk Medication Policy.
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Management manual -Aseptic Technique Policy.
- Policy Procedure Matching – Sir Charles Gairdner Hospital
- Policy- Procedure Matching – Osborne Park Hospital
- SCGOPHCG Clinical Handover - Communicating for Safety
- SCGOPHCG Nursing Practice Guideline 26 Radiological Examinations
- SCGH Urology Departmental Guideline IV Fluid Therapy in Post - Obstruction Polyuria

Patient Education Resource:

- SCGH Urology Department- Care of your Nephrostomy Tube – Patient Information

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https://www.cookmedical.com/data/IFU_PDF/T_CMLC_REV2.PDF (Product information)

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<https://cloud.merit.com/catalog/Brochures/403117001-A.pdf> (Product information)

Authorisation & Version Control

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